

Duality Theory Notes

Math Camp 2026 — Teaching Notes

Primal: pick inputs to minimize cost while meeting output targets.
Dual: find the best lower bound on that minimum using shadow prices on outputs.

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1 Why duality?

Many economic problems are **cost-minimization** problems: a firm chooses how much labor, material, and other inputs to use, pays for them, and must produce at least given amounts of each output. Before computing an optimum, ask a simpler question: how low could cost *possibly* be, given only the technology and the output targets?

Key idea. Before solving exactly, ask: *what is the best lower bound we can prove on minimum cost?* Duality turns that bound into a **second optimization problem** whose variables are **shadow prices** on the outputs.

The cast of characters.

- **Inputs** (labor hours, tons of material, machine time): what the firm uses and pays for.
- **Input costs:** the dollar price per unit of each input.
- **Output targets:** minimum amounts of each good that must be produced.
- **Technology:** how much of each output one unit of each input creates.
- **Shadow prices on outputs:** internal dollar values assigned to each good in a valuation exercise (not necessarily market prices).

Worked example (two inputs, two outputs). Labor costs \$3 per hour; material costs \$5 per ton. Targets: at least 4 widgets and 6 gadgets. Technology (rows: widgets, gadgets; columns: labor per hour, material per ton):

$$\begin{pmatrix} 1 & 0 \\ \frac{1}{2} & 2 \end{pmatrix}$$

Input costs, output targets, and one feasible input mix:

$$\text{costs} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \quad \text{targets} = \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad \text{plan} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} \quad (4 \text{ labor-hours, } 2 \text{ tons of material}).$$

Total cost: $4 \times 3 + 2 \times 5 = 22$. Widgets produced: $4 \times 1 + 2 \times 0 = 4$. Gadgets produced: $4 \times \frac{1}{2} + 2 \times 2 = 6$. The **planner** asks: what is the cheapest feasible input mix? The **accountant** asks: what is the highest total value we can assign to the output targets without making any input look like a bargain at those internal output prices?

Connecting output value to a cost floor (intuition only). The accountant assigns a dollar value to each output type (widget price, gadget price) and defines

$$V = (\text{widget price}) \times 4 + (\text{gadget price}) \times 6,$$

the paper value of the *targets* (4 widgets and 6 gadgets). The claim we want is: **every feasible plan has real input cost at least V .**

To get there, compare three numbers for **any** feasible production plan:

Real input bill	what you actually pay for labor and material
↓ Link 1	
Accounting value produced	widget price $\times$ widgets made + gadget price $\times$ gadgets made
↓ Link 2	
Value of targets (V)	widget price $\times$ 4 + gadget price $\times$ 6

Link 1: $\text{real input bill} \geq \text{accounting value of what was produced}$. You pay for inputs; those inputs create outputs; each output has an accounting price. **Link 1** says the total bill cannot be *below* the accounting value of what came out. It holds only if output prices are **consistent**:

check each input separately. One hour of labor costs \$3; at your internal prices it creates $1 \times (\text{widget price}) + \frac{1}{2} \times (\text{gadget price})$ of value. If that exceeded \$3, labor would be a money machine—so consistent prices rule that out. Same check for material (\$5 per ton). Add over all inputs used.

Link 2: accounting value of what was produced $\geq V$. The plan meets the targets (at least 4 widgets and 6 gadgets), and output prices are nonnegative. So the accounting value of what you *actually* produced is at least the accounting value of the *minimum required* amounts—that is exactly V .

Chain. Link 1 and Link 2 together give

$$\text{real input bill} \geq \text{accounting value produced} \geq V.$$

So every feasible plan costs at least V ; the true minimum cannot be below V . **Duality** asks the accountant to raise V as high as possible while keeping prices consistent. At the optimum, planner and accountant agree on the number.

(If this story does not click yet, that is fine—Section 2 gives the formal math. You can treat this section as motivation only.)

The next section makes this precise for linear programs: we introduce notation, derive the primal–dual pair, prove that dual values are valid lower bounds, and show that the best dual value equals the primal optimum.

2 Linear programming duality

We build LP duality in four steps. First, derive the dual from a lower-bound argument. Second, prove **weak duality** (dual $\leq$ primal). Third, prove **strong duality** (dual = primal), using Farkas’ Lemma as the key tool. Fourth, read optimality through **complementary slackness** and extend the recipe to general LPs.

The primal–dual pair. The **primal** in input–output form is

$$(P) \quad \min \mathbf{c}^\top \mathbf{x} \quad \text{s.t.} \quad A\mathbf{x} \geq \mathbf{b}, \mathbf{x} \geq \mathbf{0}. \quad (1)$$

We now turn the informal lower-bound idea into an explicit dual problem.

Claim 1 (Dual of the standard LP). *Problem (2) is the best LP lower bound on the optimum of (1) obtained by nonnegative shadow prices on the constraints:*

$$(D) \quad \max \mathbf{b}^\top \mathbf{y} \quad \text{s.t.} \quad A^\top \mathbf{y} \leq \mathbf{c}, \mathbf{y} \geq \mathbf{0}. \quad (2)$$

Proof. Pick any $\mathbf{y} \geq \mathbf{0}$. From $A\mathbf{x} \geq \mathbf{b}$, multiplying row i by $y_i \geq 0$ and summing gives $\mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top \mathbf{y}$. If also $A^\top \mathbf{y} \leq \mathbf{c}$, then for every $\mathbf{x} \geq \mathbf{0}$,

$$\mathbf{c}^\top \mathbf{x} \geq \mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top \mathbf{y}.$$

So $\mathbf{b}^\top \mathbf{y}$ is a lower bound on primal cost for every feasible $\mathbf{x}$. Maximising this bound over $\mathbf{y}$ gives (2). $\square$

Economics. Read the pair as follows.

- x_j : quantity of input j (labor, material); c_j : its unit cost.
- $(A\mathbf{x})_i \geq b_i$: output of type i meets the target.
- y_i : **shadow price** of output i .

- $A^T \mathbf{y} \leq \mathbf{c}$: value of outputs attributable to each input $\leq$ its cost.
- $\mathbf{b}^T \mathbf{y}$: total value of output targets at prices $\mathbf{y}$.

A small illustration fixes the notation (one output type, so A is a single row). Consider $\min 3x_1 + 5x_2$ s.t. $x_1 + 2x_2 \geq 4$, $x_1, x_2 \geq 0$, i.e. $A = (1 \ 2)$: one unit of labor yields $a_{11} = 1$ unit of output, one unit of material yields $a_{12} = 2$ units. ($x_1 = \text{labor}$, $x_2 = \text{material}$; produce at least 4 units at minimum cost). The dual is $\max 4y$ s.t. $y \leq 3$, $2y \leq 5$, $y \geq 0$, which gives $y^* = 3$ and cost $= 12$.

Having defined the pair, the first theorem says that every feasible dual solution gives a valid bound.

Weak duality.

Theorem 1 (Weak duality). *For every feasible $\mathbf{x}$ of (1) and $\mathbf{y}$ of (2), $\mathbf{c}^T \mathbf{x} \geq \mathbf{b}^T \mathbf{y}$, hence $p^* \geq d^*$.*

Proof. Dual feasibility gives $A^T \mathbf{y} \leq \mathbf{c}$. Since $\mathbf{x} \geq \mathbf{0}$, we get $\mathbf{c}^T \mathbf{x} \geq \mathbf{y}^T A \mathbf{x}$. Primal feasibility gives $A \mathbf{x} \geq \mathbf{b}$; since $\mathbf{y} \geq \mathbf{0}$, $\mathbf{y}^T A \mathbf{x} \geq \mathbf{b}^T \mathbf{y}$. Chaining the two inequalities yields $\mathbf{c}^T \mathbf{x} \geq \mathbf{b}^T \mathbf{y}$. Taking the minimum over primal feasible $\mathbf{x}$ and the maximum over dual feasible $\mathbf{y}$ gives $p^* \geq d^*$. $\square$

Remark 2 (Optimality certificate). *If feasible $\mathbf{x}, \mathbf{y}$ satisfy $\mathbf{c}^T \mathbf{x} = \mathbf{b}^T \mathbf{y}$, both are optimal.*

Economics. Input cost is always at least total output value. Equality is a **certificate of efficiency**: the input mix and output prices are mutually consistent.

Weak duality gives $d^* \leq p^*$, but not yet $d^* = p^*$. To close the gap we need a theorem of alternatives; Farkas' Lemma is the right tool.

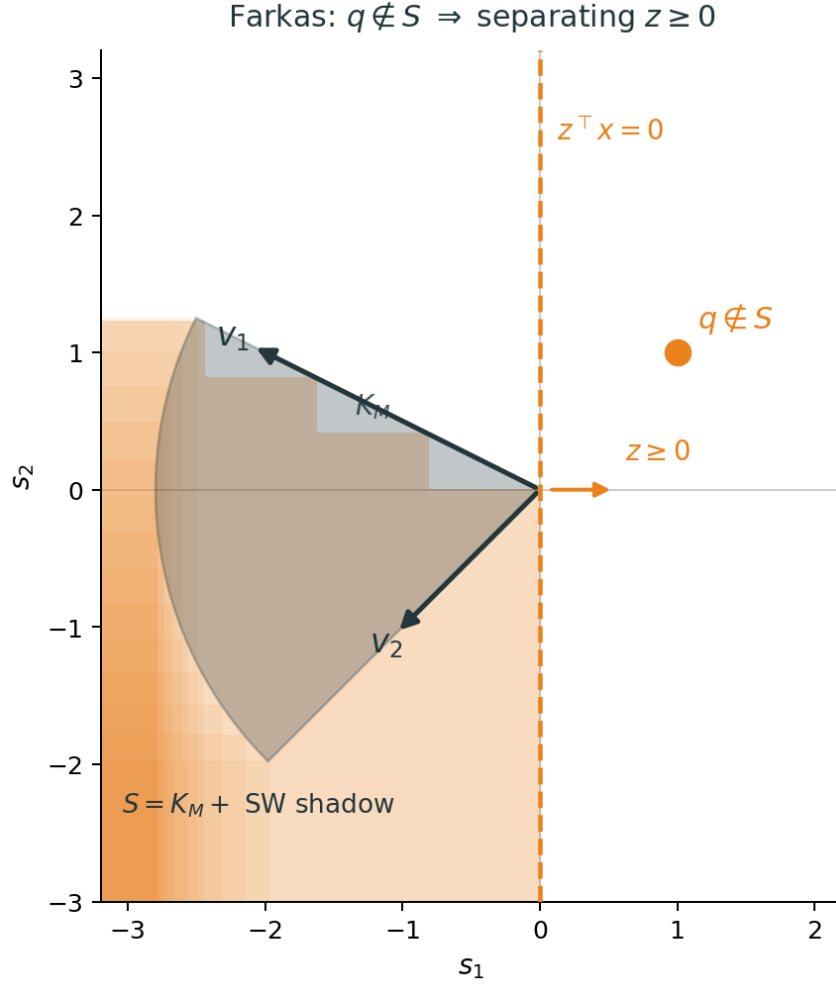
Farkas' Lemma.

Theorem 3 (Farkas' Lemma — inequality form). *Let $M \in \mathbb{R}^{m \times n}$, $\mathbf{q} \in \mathbb{R}^m$. Exactly one of the following has a solution:*

- (i) $\exists \mathbf{x} \geq \mathbf{0}$ with $M \mathbf{x} \geq \mathbf{q}$;
- (ii) $\exists \mathbf{z} \geq \mathbf{0}$ with $M^T \mathbf{z} \leq \mathbf{0}$ and $\mathbf{q}^T \mathbf{z} > 0$.

The lemma has a clean geometric reading. Let $K_M = \{M \mathbf{x} : \mathbf{x} \geq \mathbf{0}\}$ be the column cone of M , and let $S = K_M + (-\mathbb{R}_+^m) = \{M \mathbf{x} - \mathbf{y} : \mathbf{x}, \mathbf{y} \geq \mathbf{0}\}$, the cone plus free disposal in every negative coordinate direction.

Claim 2 (Geometric form of Farkas). *Alternative (i) holds if and only if $\mathbf{q} \in S$. Alternative (ii) holds if and only if $\mathbf{q} \notin S$; then a separating normal $\mathbf{z} \geq \mathbf{0}$ certifies that $\mathbf{q}$ lies strictly above the hyperplane $\mathbf{z}^T \mathbf{s} = 0$.*



Proof of Farkas' Lemma. Step 1 (at most one alternative). Suppose both (i) and (ii) hold. Pick $\mathbf{x} \geq \mathbf{0}$ with $M\mathbf{x} \geq \mathbf{q}$ and $\mathbf{z} \geq \mathbf{0}$ with $M^T\mathbf{z} \leq \mathbf{0}$ and $\mathbf{q}^T\mathbf{z} > 0$. Then

$$0 < \mathbf{q}^T\mathbf{z} \leq (M\mathbf{x})^T\mathbf{z} = \mathbf{x}^T(M^T\mathbf{z}) \leq 0,$$

a contradiction.

Step 2 (if (i) fails, then (ii) holds). If (i) fails, then $\mathbf{q} \notin S$. Since S is convex and closed, there exists $\mathbf{z} \neq \mathbf{0}$ with $\mathbf{z}^T\mathbf{s} \leq 0$ for all $\mathbf{s} \in S$ and $\mathbf{z}^T\mathbf{q} > 0$. For each standard basis vector $\mathbf{e}_i$, the point $-\mathbf{e}_i \in S$ (take $\mathbf{x} = \mathbf{0}$, $\mathbf{y} = \mathbf{e}_i$), so $-z_i \leq 0$, i.e. $z_i \geq 0$. For each $\mathbf{x} \geq \mathbf{0}$, the point $M\mathbf{x} \in S$ (take $\mathbf{y} = \mathbf{0}$), so $\mathbf{z}^TM\mathbf{x} \leq 0$, i.e. $M^T\mathbf{z} \leq \mathbf{0}$. Thus (ii) holds. $\square$

Economics. Either a target is feasible with free disposal, or a nonnegative price vector exposes it as too ambitious—a no-arbitrage statement.

With Farkas in hand, we can show that no strictly better dual bound is possible than the primal optimum.

Strong duality.

Theorem 4 (Strong duality for LP). *If (1) has an optimal value p^* , then (2) has $d^* = p^*$.*

Proof. Fix $\varepsilon > 0$. Because p^* is optimal, no $\mathbf{x} \geq \mathbf{0}$ satisfies $A\mathbf{x} \geq \mathbf{b}$ and $\mathbf{c}^T\mathbf{x} \leq p^* - \varepsilon$. Equivalently, the system

$$A\mathbf{x} \geq \mathbf{b}, \quad \mathbf{c}^T\mathbf{x} \leq p^* - \varepsilon, \quad \mathbf{x} \geq \mathbf{0}$$

is infeasible. Write this as $M\mathbf{x} \geq \mathbf{q}$ with $M = \begin{pmatrix} A \\ -\mathbf{c}^\top \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} \mathbf{b} \\ -(p^* - \varepsilon) \end{pmatrix}$. By Farkas, (ii) holds: $\exists \mathbf{y} \geq \mathbf{0}, \mu \geq 0$ with $A^\top \mathbf{y} - \mu \mathbf{c} \leq \mathbf{0}$ and $\mathbf{b}^\top \mathbf{y} - \mu(p^* - \varepsilon) > 0$. If $\mu = 0$, then $\mathbf{b}^\top \mathbf{y} > 0$ while any primal feasible $\mathbf{x}$ gives $\mathbf{b}^\top \mathbf{y} \leq \mathbf{y}^\top A\mathbf{x} \leq \mathbf{c}^\top \mathbf{x}$, contradicting primal feasibility; hence $\mu > 0$. Divide by μ and set $\tilde{\mathbf{y}} = \mathbf{y}/\mu$. Then $A^\top \tilde{\mathbf{y}} \leq \mathbf{c}$ and $\mathbf{b}^\top \tilde{\mathbf{y}} > p^* - \varepsilon$, so $\tilde{\mathbf{y}}$ is dual feasible with objective exceeding $p^* - \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $d^* \geq p^*$. Together with weak duality, $d^* = p^*$. $\square$

Economics. Minimum input cost = maximum output value at optimality. Efficient production can be supported by shadow output prices.

Strong duality identifies the optimum value; complementary slackness identifies *which* constraints matter at the optimum.

Complementary slackness.

Theorem 5 (Complementary slackness). *Feasible $\mathbf{x}, \mathbf{y}$ are optimal iff*

$$(A\mathbf{x} - \mathbf{b})_i y_i = 0 \quad \forall i, \quad (\mathbf{c} - A^\top \mathbf{y})_j x_j = 0 \quad \forall j.$$

In words:

If ...	then ...
primal constraint i is slack $y_i > 0$	$y_i = 0$ primal constraint i is tight
dual constraint j is slack $x_j > 0$	$x_j = 0$ dual constraint j is tight

Necessity. Suppose $\mathbf{x}, \mathbf{y}$ are optimal. By strong duality, $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$. In the weak-duality chain,

$$\mathbf{c}^\top \mathbf{x} \geq \mathbf{y}^\top A\mathbf{x} \geq \mathbf{b}^\top \mathbf{y},$$

both inequalities must be equalities. Hence $\mathbf{y}^\top (A\mathbf{x} - \mathbf{b}) = 0$ and $(\mathbf{c} - A^\top \mathbf{y})^\top \mathbf{x} = 0$. Each term in these dot products is a product of nonnegative factors, so every term must be zero. $\square$

Sufficiency. If complementary slackness holds, then

$$\mathbf{c}^\top \mathbf{x} = \mathbf{y}^\top A\mathbf{x} + (\mathbf{c} - A^\top \mathbf{y})^\top \mathbf{x} = \mathbf{y}^\top A\mathbf{x} = \mathbf{b}^\top \mathbf{y} + \mathbf{y}^\top (A\mathbf{x} - \mathbf{b}) = \mathbf{b}^\top \mathbf{y}.$$

Weak duality then certifies that both are optimal. $\square$

Remark 6 (Direct proof). *From $\mathbf{c}^\top \mathbf{x} = \mathbf{b}^\top \mathbf{y}$ and weak duality: any feasible $\tilde{\mathbf{x}}$ has $\mathbf{c}^\top \tilde{\mathbf{x}} \geq \mathbf{c}^\top \mathbf{x}$; any feasible $\tilde{\mathbf{y}}$ has $\mathbf{b}^\top \tilde{\mathbf{y}} \leq \mathbf{b}^\top \mathbf{y}$.*

Economics. Zero-profit / full-utilization logic. Output target slack $\Rightarrow$ zero shadow price; binding output target $\Rightarrow$ positive price. Input not used $\Rightarrow x_j = 0$; input used $\Rightarrow$ breaks even at shadow output prices. Links to the Day 3 producer-theory slide (efficient production supported by prices when free disposal holds).

The derivation above used $\geq$ constraints and $\mathbf{x} \geq \mathbf{0}$. The same bounding identity works for any mix of constraint and variable types; only the signs change.

Dual of a general LP.

Key idea. Same bounding argument; signs follow from constraint and variable types.

Claim 3 (Dual rules for a general LP). *Consider a primal **minimisation** with any mix of $\geq$, $\leq$, $=$ constraints and nonnegative, nonpositive, or free variables. The dual is a **maximisation** with one dual variable y_i per primal constraint. If the primal is a **maximisation**, reverse every inequality in the table below.*

Proof. For any primal feasible $\mathbf{x}$ and dual candidate $\mathbf{y}$,

$$\mathbf{c}^\top \mathbf{x} - \mathbf{b}^\top \mathbf{y} = (\mathbf{c} - A^\top \mathbf{y})^\top \mathbf{x} + \mathbf{y}^\top (A\mathbf{x} - \mathbf{b}).$$

Require each bracketed term to be ≥ 0 for *all* primal feasible $\mathbf{x}$. The sign of y_i is pinned down by the type of primal constraint i ; the type of dual constraint on $(A^\top \mathbf{y})_j$ is pinned down by the sign of x_j . Maximising $\mathbf{b}^\top \mathbf{y}$ subject to these conditions gives the dual. The explicit rules for a primal minimum (dual maximum) are:

Primal	Dual
$\mathbf{a}_i^\top \mathbf{x} \geq b_i$	$y_i \geq 0$
$\mathbf{a}_i^\top \mathbf{x} \leq b_i$	$y_i \leq 0$
$\mathbf{a}_i^\top \mathbf{x} = b_i$	y_i free
$x_j \geq 0$	$(A^\top \mathbf{y})_j \leq c_j$
$x_j \leq 0$	$(A^\top \mathbf{y})_j \geq c_j$
x_j free	$(A^\top \mathbf{y})_j = c_j$

□

For example,

$$\min 3x_1 - 2x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 5, \quad 2x_1 - x_2 \leq 10, \quad x_1 - 3x_2 = 4, \quad x_1 \geq 0, \quad x_2 \text{ free}$$

has dual

$$\max 5y_1 + 10y_2 + 4y_3 \quad \text{s.t.} \quad y_1 + 2y_2 + y_3 \leq 3, \quad y_1 - y_2 - 3y_3 = -2, \quad y_1 \geq 0, \quad y_2 \leq 0, \quad y_3 \text{ free.}$$

Section 3 shows that this entire LP story is one instance of a broader Lagrangian construction.

3 Lagrangian duality

LP duality multiplies constraints by nonnegative weights, adds them to the objective, and maximises the resulting lower bound. The same recipe applies to nonlinear problems. We proceed as follows: state the general primal and dual; prove weak duality; interpret the dual geometrically in two settings; recover the LP dual from Section 2; and close with strong duality under convexity and Slater's condition.

The general primal is

$$(P) \quad \min f_0(\mathbf{x}) \quad \text{s.t.} \quad f_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, m, \quad h_j(\mathbf{x}) = 0, \quad j = 1, \dots, p, \quad \mathbf{x} \in D,$$

with optimal value p^* . The **Lagrangian** attaches multipliers to each constraint:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_{i=1}^m \lambda_i f_i(\mathbf{x}) + \sum_{j=1}^p \nu_j h_j(\mathbf{x}), \quad \boldsymbol{\lambda} \geq \mathbf{0}, \quad \boldsymbol{\nu} \text{ free.}$$

For any feasible $\mathbf{x}$, each added term is ≤ 0 , so $L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) \leq f_0(\mathbf{x})$. The **dual function** minimises the Lagrangian over $\mathbf{x}$:

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) = \inf_{\mathbf{x} \in D} L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}).$$

The **dual problem** chooses multipliers to sharpen this bound:

$$(D) \quad \max g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \text{ s.t. } \boldsymbol{\lambda} \geq \mathbf{0}, \quad d^* = \sup_{\boldsymbol{\lambda} \geq \mathbf{0}} g(\boldsymbol{\lambda}, \boldsymbol{\nu}).$$

Theorem 7 (Weak duality for the Lagrangian dual). *Always $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$ for $\boldsymbol{\lambda} \geq \mathbf{0}$, hence $d^* \leq p^*$. The quantity $p^* - d^*$ is the **duality gap**.*

Proof. For any primal feasible $\mathbf{x}$ and any $\boldsymbol{\lambda} \geq \mathbf{0}$, $\boldsymbol{\nu}$,

$$g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq L(\mathbf{x}, \boldsymbol{\lambda}, \boldsymbol{\nu}) = f_0(\mathbf{x}) + \sum_i \lambda_i f_i(\mathbf{x}) + \sum_j \nu_j h_j(\mathbf{x}) \leq f_0(\mathbf{x}),$$

because $f_i(\mathbf{x}) \leq 0$, $h_j(\mathbf{x}) = 0$, and $\lambda_i \geq 0$. Taking the infimum over primal feasible $\mathbf{x}$ on the right gives $g(\boldsymbol{\lambda}, \boldsymbol{\nu}) \leq p^*$. Maximising over $\boldsymbol{\lambda}, \boldsymbol{\nu}$ yields $d^* \leq p^*$. $\square$

Pictures make the dual function easy to remember: it is the intercept of the best supporting line or hyperplane below the cloud of all achievable constraint-objective pairs.

Geometry I: one inequality and the (u, t) cloud. Key idea. Dual = best lower bound from a supporting **line** below the whole achievable set $\mathcal{A} \subset \mathbb{R}^2$.

Consider $\min_{\mathbf{x}} f_0(\mathbf{x})$ s.t. $f_1(\mathbf{x}) \leq 0$. Write $u = f_1(\mathbf{x})$ and $t = f_0(\mathbf{x})$. As $\mathbf{x}$ runs over D , the pairs

$$\mathcal{A} = \{ (u, t) : u = f_1(\mathbf{x}), t = f_0(\mathbf{x}), \mathbf{x} \in D \}$$

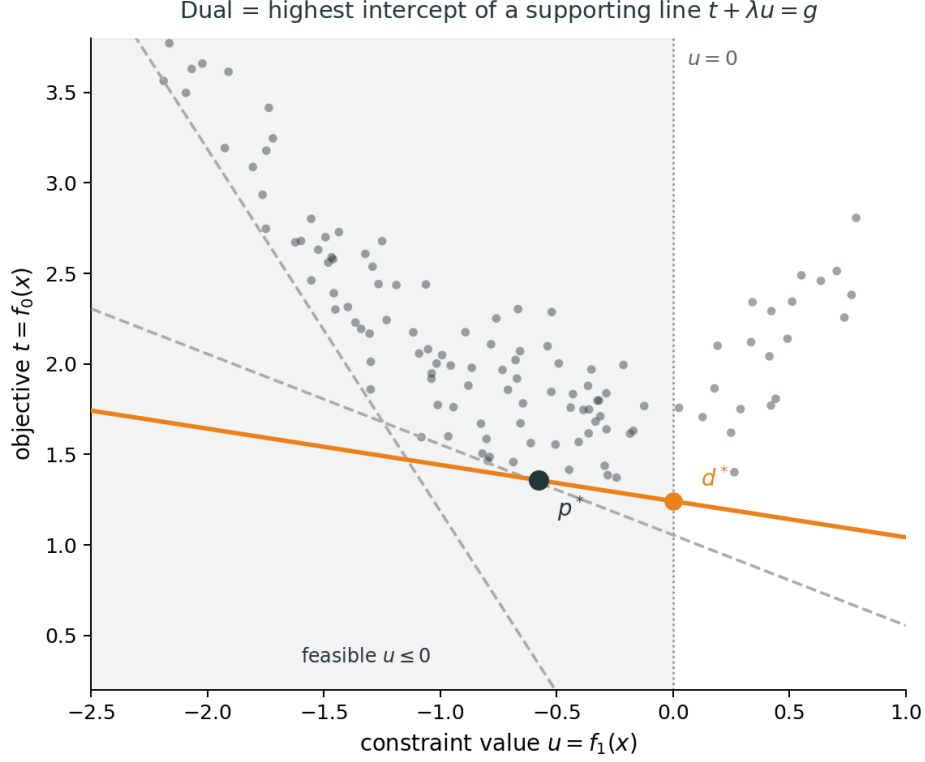
form the **whole cloud** of achievable outcomes, both feasible ($u \leq 0$) and infeasible. The primal value p^* is the smallest t among points in $\mathcal{A}$ with $u \leq 0$. For $\lambda \geq 0$, lines $t + \lambda u = \alpha$ have slope $-\lambda$ in the (u, t) plane, and

$$g(\lambda) = \inf_{(u, t) \in \mathcal{A}} (t + \lambda u)$$

is the smallest α such that the line still touches $\mathcal{A}$ from below (its intercept on $u = 0$). The infimum is over **all** of $\mathcal{A}$, not only the feasible slice $u \leq 0$.

Claim 4 (Geometric weak duality in the (u, t) picture). *For every $\lambda \geq 0$, $g(\lambda) \leq p^*$. The dual value $d^* = \sup_{\lambda \geq 0} g(\lambda)$ is the largest intercept at $u = 0$ obtained by a supporting line below $\mathcal{A}$.*

Proof. Take any feasible point $(u, t) \in \mathcal{A}$ with $u \leq 0$. Since $\lambda \geq 0$, $t + \lambda u \leq t$. Also $g(\lambda) \leq t + \lambda u$ by definition of the infimum. Hence $g(\lambda) \leq t$. Minimising t over feasible points gives $g(\lambda) \leq p^*$. Maximising $g(\lambda)$ over $\lambda \geq 0$ gives d^* . $\square$



Geometry II: one inequality, one equality, and the (u, v, t) cloud. Key idea. Adding an equality lets the supporting object tilt in a **third** direction; free ν can tighten the bound.

Now add $h(\mathbf{x}) = 0$ and write $u = f_1(\mathbf{x})$, $v = h(\mathbf{x})$, $t = f_0(\mathbf{x})$. The cloud

$$\mathcal{A} = \{ (u, v, t) : u = f_1(\mathbf{x}), v = h(\mathbf{x}), t = f_0(\mathbf{x}), \mathbf{x} \in D \}$$

again contains **all** achievable triples; feasible points satisfy $u \leq 0$ and $v = 0$. Among them, the primal minimises t , giving p^* . Choose $\lambda \geq 0$ (for u) and ν free (for v). The plane $t + \lambda u + \nu v = \alpha$ supports $\mathcal{A}$ from below, and

$$g(\lambda, \nu) = \inf_{(u, v, t) \in \mathcal{A}} (t + \lambda u + \nu v)$$

is its intercept at $u = 0$, $v = 0$.

Claim 5 (Geometric weak duality in the (u, v, t) picture). *For every $\lambda \geq 0$ and ν , $g(\lambda, \nu) \leq p^*$. The dual problem $d^* = \sup_{\lambda \geq 0, \nu} g(\lambda, \nu)$ seeks the best intercept from a hyperplane below the full cloud $\mathcal{A}$.*

Proof. For any feasible (u, v, t) , $u \leq 0$ and $v = 0$, so $t + \lambda u + \nu v = t + \lambda u \leq t$. Since $g(\lambda, \nu) \leq t + \lambda u + \nu v$, we get $g(\lambda, \nu) \leq p^*$. $\square$

Remark 8 (Why $\nu = 0$ can be weaker). *With $\nu = 0$, the plane $t + \lambda u = \alpha$ cannot tilt in the v -direction. The infimum $g(\lambda, 0)$ may occur at $v \neq 0$, giving a low intercept at $v = 0$. Allowing ν to vary lets the plane tilt so the minimum over $\mathcal{A}$ can occur nearer $v = 0$, often raising the bound; for convex problems this can achieve $d^* = p^*$. For LPs, $\mathcal{A}$ is a polyhedron (convex), so strong duality holds when the primal optimum is finite.*

Having seen the geometry, we verify that the LP dual of Section 2 is exactly what the Lagrangian recipe produces.

LP recovers Section 2.

Claim 6 (LP Lagrangian dual equals the LP dual). Write $\min \mathbf{c}^\top \mathbf{x}$ s.t. $A\mathbf{x} \geq \mathbf{b}$, $\mathbf{x} \geq \mathbf{0}$ as $f_0(\mathbf{x}) = \mathbf{c}^\top \mathbf{x}$, $f_1(\mathbf{x}) = \mathbf{b} - A\mathbf{x} \leq \mathbf{0}$, $f_2(\mathbf{x}) = -\mathbf{x} \leq \mathbf{0}$. The Lagrangian dual (2) is obtained with $\mathbf{y} = \boldsymbol{\lambda}$.

Proof. The Lagrangian is

$$L = \mathbf{c}^\top \mathbf{x} + \boldsymbol{\lambda}^\top (\mathbf{b} - A\mathbf{x}) + \boldsymbol{\mu}^\top (-\mathbf{x}).$$

For fixed $(\boldsymbol{\lambda}, \boldsymbol{\mu})$,

$$g(\boldsymbol{\lambda}, \boldsymbol{\mu}) = \inf_{\mathbf{x}} [(\mathbf{c} - A^\top \boldsymbol{\lambda} - \boldsymbol{\mu})^\top \mathbf{x} + \boldsymbol{\lambda}^\top \mathbf{b}].$$

If $A^\top \boldsymbol{\lambda} + \boldsymbol{\mu} \neq \mathbf{c}$, the infimum is $-\infty$. If $A^\top \boldsymbol{\lambda} + \boldsymbol{\mu} = \mathbf{c}$, then $g = \boldsymbol{\lambda}^\top \mathbf{b}$. Requiring $\boldsymbol{\lambda} \geq \mathbf{0}$ and $\boldsymbol{\mu} \geq \mathbf{0}$ and writing $A^\top \boldsymbol{\lambda} \leq \mathbf{c}$ gives (2) with $\mathbf{y} = \boldsymbol{\lambda}$. $\square$

When the problem is convex and strictly feasible, weak duality upgrades to strong duality.

Strong duality for convex problems. The problem is **convex** if f_0, f_i are convex, h_j are affine, and D is convex.

Theorem 9 (Strong duality). *If the primal is convex, **Slater's condition** holds (strict feasibility: $f_i(\mathbf{x}) < 0$, $h_j(\mathbf{x}) = 0$ for some $\mathbf{x} \in \text{relint } D$), and p^* is finite, then $d^* = p^*$ and the dual optimum is attained.*

Remark 10 (What is $\text{relint } D$?). The **relative interior** $\text{relint } D$ is the “interior” of D measured inside the flat (affine hull) that contains D , not inside all of $\mathbb{R}^n$.

- If $D = \mathbb{R}^n$, then $\text{relint } D = \mathbb{R}^n$.
- If D is a line segment in $\mathbb{R}^2$, its usual topological interior is empty, but $\text{relint } D$ is the **open segment** (endpoints excluded).
- If $D = \{\mathbf{x} : A\mathbf{x} \leq \mathbf{b}\}$ is a full-dimensional polyhedron, $\text{relint } D$ is the set of points that satisfy every inequality **strictly** ($A\mathbf{x} < \mathbf{b}$), i.e. not on a lower-dimensional face.

Slater asks for a **strictly feasible** point that is not trapped on a boundary of D : inequalities hold with slack ($f_i(\mathbf{x}) < 0$), equalities hold exactly, and $\mathbf{x}$ lies in the relative interior of the domain.

For LPs, a feasible point with finite optimum suffices. Slater is the convex analogue of “interior point exists.”

Economics. λ_i, ν_j are shadow prices on constraints. The dual finds the price system giving the best accounting lower bound on optimal cost. When the gap vanishes, prices fully explain the optimum.